



Distance Metrics

A. Given the following points in a 2-dimensional vector space:

	X	Y
p1	6	3
p2	2	2
p3	3	4

1. Compute the pairwise **Manhattan distance** between all points.
2. Compute the pairwise **Euclidean distance** between all points.
3. For each metric, identify which two points are the closest.
4. Briefly compare the two metrics:
 - What is similar?
 - What is different?

	manhattan	euclidean
p1 > p2	5	4.12 (sqr(17))
p1 > p3	4	3.16 (sqr(10))
p2 > p3	3	2.24 (sqr(5))

B. Hamming Distance

Consider the two binary feature vectors:

- a = 1010010
- b = 0011001

1. Compute the **Hamming distance** between the two vectors. hamming dist(a,b) = 4
2. Mark the positions where the two vectors differ.
3. Explain briefly why **XOR** can be used to compute the Hamming distance efficiently.
4. Write one line of Python/Numpy code that computes the Hamming distance.

3. XOR will eliminate 1s that are the same across both feature vectors, then count ones

4. `XOR(a,b).bit_count()`

C. Mixed-data challenge

You are given the following small customer dataset:

Customer	Age	Income (€)	Premium Account	Satisfaction
C1	24	32000	yes	medium
C2	26	34000	no	high
C3	52	78000	yes	low
C4	49	76000	no	medium

categorical nominal

numeric

binary

ordinal

1. Classify each feature as:

- numeric
- binary
- ordinal
- categorical nominal

2. Which features could be used directly with **Euclidean distance** in a natural way? age, income

3. Which features would require special care before applying a distance-based clustering method?

4. Suppose we encode:

- Premium Account: no = 0, yes = 1
- Satisfaction: low = 1, medium = 2, high = 3

income due to large magnitude,
suitable encoding for premium and
satisfaction

Discuss briefly:

- Which of these encodings is more defensible?
- Which one is more problematic?
- Why?

premium acc more defensible than satisfaction
since it signifies a presence/absence of smth
satisfaction raises question of scale, why start at 1
and not 0, why the given distances between the values?

5. Compute the **Euclidean distance** between C1 and C2 using only the numeric features:

- Age
- Income

6. Why would this distance ignore potentially relevant information about the customers?

euclidean distance

$C1 > C2 \quad \sqrt{(26-24)^2 + (34000-32000)^2} \sim 2000.0$
> scaling problem, income totally dominates age

output does not take into account premium and satisfaction,
which going by the "customer" signifier is smth that should be evaluated

in this case, the customers have similar age and income but different
experiences as customers > differing strongly in important behavioral and attitudinal characteristics

overall, however, this is a matter of what exactly should be accomplished by the resulting model